01 Vega Health Model Monitoring Plan

For each Use Case, Vega Health will construct a monitoring plan following the process outlined in this document. Based on the selected Use Case and key dimensions of the in-scope AI product, Vega Health will guide the client through a series of decision points. Inputs from these decision points will determine the monitoring metrics, alerts, and accountable individuals for each Use Case.

Vega Health maintains a library of metrics, tests, and statistical analyses that address each dimension of model monitoring. While the exact combination of metrics may vary for different AI products, each monitoring dashboard – as well as the associated workflows and alerts – is intended to equip health system stakeholders with the information necessary to manage the lifecycle of the Use Case

The Vega Health Model Monitoring Plan will be fully completed and delivered to client leadership for approval for each in-scope Use Case. ‘

02 Use Case Summary

> *Vega Health will work with the Sponsors for each Use Case to compile a Use Case Summary. The Use Case Summary provides baseline context for the Monitoring Plan, helps ascribe key stakeholders and accountable parties for monitoring workflows, and ensures monitoring metrics are actionable and informative.*

Exhibit 2.1. Use Case Summary & Model Details

Key Details	Description
Intended Use	*One or two sentences: who uses the output, for which patients or tasks, to make what decision or do what work, at what point in the workflow.*
Problem Addressed	*What happens today and why it isn't good enough.*
Workflow & Action	*What the solution produces, who sees it, where (e.g. EHR advisory, worklist, draft note, inbox draft), and what they are expected to do with it.*
Model	*Developer or vendor, model type, training population (predictive) or underlying foundation model (generative), version.*
Go-Live Details	*Date, sites, units, and user groups in scope.*
Monitoring Objective	*Default: "Confirm, continuously and through periodic review, that the solution remains technically reliable, performs within agreed bounds, is used as intended, and is associated with the expected outcomes; detect problems early enough to act."*

03 Key Decision Points

> *Key Decision Points represent the inputs that determine which set of monitoring metrics are most appropriate for the specific Use Case.*

Exhibit 3.1. Key Decision Points

#	Decision	Options	Effect on the Monitoring Plan
P1	Modality	Predictive / Generative	Selects the *[Predictive]* or *[Generative]* blocks in §4, §5, and §6, and the suggested core set
P2	*[Predictive]* Output and action	Drives alerts to users / score or tier displayed / batch worklist	**Alerts:** alert-driven core set (alert-episode accuracy and alert burden). **Displayed or worklist:** not-alert-driven core set (operating point or calibration, engagement)
P3	*[Predictive]* Outcome labels	Label source; label latency; available in the Platform's feeds?	Sets when label-based metrics first report and their lookback windows (no shorter than the latency). **Not available:** label-based Model Performance metrics are dropped and Model Performance relies on drift and stability; state this in §3
P4	*[Generative]* Task type	Clinical documentation drafting / summarization / patient-message drafting / extraction or coding / question answering or retrieval / other	Sets the review rubric dimensions (§5.4), the significant-error definitions for Clinically significant error rate (PERF-19), the safety screens in Safety event rate (PERF-23), and the primary outcome (§4)
P5	*[Generative]* Human in the loop	Clinician reviews every output before use / output used without routine review / output reaches patients	Sets severity and review intensity. **Without routine review or patient-facing:** Clinically significant error rate (PERF-19) and Safety event rate (PERF-23) are critical-severity, review sample size is increased, and Safety event rate (PERF-23) moves into the core set
P6	*[Generative]* Edit capture	Edits to outputs are captured / not captured	**Captured:** Draft acceptance / edit rate (ADPT-08) is in the core set and is the Adoption & Workflow primary metric. **Not captured:** Engagement (ADPT-05) is primary
P7	*[Generative]* Quality review mode	Human review / automated review / both	**Human:** Human-rated output quality (PERF-18) and Clinically significant error rate (PERF-19), with a customer reviewer commitment (§10). **Automated:** Automated quality score (PERF-20) with periodic Judge–human agreement (PERF-21) on a small human-reviewed sample. **Both:** automated weekly, human monthly or quarterly

#	Decision	Options	Effect on the Monitoring Plan
P8	Update pattern	Locked, vendor-versioned / hosted foundation model updated by its provider	Sets the version-change protocol (§9). **Hosted:** Model version conformance (SYS-10) is in the core set and the provider's version is pinned
P9	Baseline source	Default: label-free metrics from the first *[4–8]* weeks of production; label-based and quality floors from vendor-reported or published performance; optional historical pre-period for Impact & ROI	Sets §6. Thresholds referenced to vendor figures are provisional until confirmed against local data at the first quarterly review
P10	Impact & ROI design	Historical pre-period comparison / trend only / not monitored (with reason)	**Pre-period:** client provides *[12]* months of pre-go-live outcome data. **Trend only:** reported as trends, without attribution. **Not monitored:** Impact & ROI tab shows n/a with the reason
P11	Equity strata	Library defaults / defaults plus additions / defaults minus removals (with reason)	Sets the strata in §5.2 and the stratified metrics

04 Model Monitoring Capabilities & Limitations

> *Depending on the specific Use Case dimensions and the inputs provided as Key Decision Points, model monitoring may rely on different metrics and tests – each with different interpretations and limitations. This section highlights some limitations of model monitoring for specific combinations of model dimensions.*

No counterfactual. With no silent evaluation or concurrent comparison, monitoring shows how the AI product performs and what happens while it is in use, not what would have happened without it. Impact & ROI results are trends, or before/after comparisons against a historical pre-period (P10); they are not proof that the AI product caused the change.

[Predictive] **Accuracy is measured on outcomes the AI product may influence.** When users act on the output, they can prevent the outcome it predicts, so measured accuracy can fall even when the model is working. Label-based accuracy metrics are therefore reported as trends against their baseline and investigated when they move; they are not a fixed measure of model quality.

[Predictive] **Labels lag.** Label-based metrics report only once outcomes are observable, *[label latency]* after the prediction.

[Generative] **Quality comes from a sample.** Human review rates a sample of *[n]* outputs per *[month]*; results carry sampling error, shown as confidence intervals. Rare failure types may not appear in any given sample, which is why safety screening and user reporting run alongside review.

[Generative] **Automated scores are only as good as their agreement with clinicians.** Where automated quality score (PERF-20) is used, it is trusted only while judge–human agreement (PERF-21) stays above its floor.

Baselines start provisional. Until local baselines are set, thresholds referenced to vendor figures may need adjustment.

05 Baseline & Alert Routing [Accountability]

> *In order to effectively implement model monitoring, stakeholders must understand expected local performance and establish processes to respond when a monitored metric varies from the expected or acceptable bounds.*

Exhibit 5.1. Baseline Construction Methodology

Baseline	Source	Window [date set]	Metrics referenced	Status
Label-free reference distributions	First *[6]* weeks of production	*To be defined*	Input record volume (SYS-03), drift and stability metrics	*Provisional until set*
Label-based or quality floors	*Vendor-reported or published performance (cite)*	*To be defined*		*Provisional until the first quarterly review*
Impact pre-period (if P10)	*Customer historical data, [12] months before go-live*	*To be defined*	Primary outcome (IMPT-01)	

Re-baselining. Baselines are reset after a model version change, a material data source or mapping change, or a formal evaluation (Framework: *Baselines and re-baselining*). Each reset is recorded with its date and reason.

Exhibit 5.2. Alert Routing Logic & Owners

Severity	Notified	Channel	Response time	Expected response
Critical	*[Metric owner; Vega support; Use Case Sponsor]]*	*[Page / email / ticket]*	*[4 business hours]*	*Investigate; consider suspension (§9)*
Warning	*[Metric owner]*	*[Email / ticket]*	*[5 business days]*	*Investigate; record the finding in the next monthly summary*
Informational	*[Metric owner]*	*Included in the monthly summary*	—	*Review at the next scheduled review*

06 Dashboard: Monitoring Dimensions & Primary Metrics

> *The primary monitoring dashboard always has four dimensions [e.g., System, Model Performance, Adoption & Workflow, Impact & ROI]. In production, all four dimensions are active [Impact & ROI is unavailable in cases where Key Decision Point P10 is not monitored]. Each dimension displays the primary metric first, along with additional monitoring metrics selected from the Vega Health monitoring Metric Library.*

Exhibit 6.1. Primary Metrics by Monitoring Dimension

Dimension	*[Predictive, alert-driven]*	*[Predictive, not alert-driven]*	*[Generative]*	Headline value	Trend chart
System	Scoring availability (SYS-01)	Scoring availability (SYS-01)	Scoring availability (SYS-01)	Trailing 7-day rate	Daily, last 28 days
Model Performance	Alert-episode PPV (PERF-02) and Encounter sensitivity (PERF-03)	Operating point (PERF-01), or Calibration (PERF-06) if there is no threshold	**Human review:** Human-rated output quality (PERF-18) and Clinically significant error rate (PERF-19). **Automated:** Automated quality score (PERF-20) with Judge–human agreement	Trailing *[90]* days with CI (generative automated: trailing 7 days)	Monthly *(generative automated: weekly)*

Dimension	*[Predictive, alert-driven]*	*[Predictive, not alert-driven]*	*[Generative]*	Headline value	Trend chart
			(PERF-21) status		
Adoption & Workflow	Alert burden (ADPT-01)	Engagement (ADPT-05)	Draft acceptance / edit rate (ADPT-08) if P6 = captured; otherwise Engagement (ADPT-05)	Trailing 4 weeks	Weekly
Impact & ROI	Primary outcome (IMPT-01): *[name the outcome]*	Primary outcome (IMPT-01): *[name the outcome]*	Primary outcome (IMPT-01): *[name the outcome, e.g. documentation time per encounter]*	Trailing 90 days	Monthly, with go-live marker and pre-period reference (P10)

07 Metric Library

> *Vega Health's monitoring Metric Library is the complete set of metrics available for each type of model Use Case. Clients can choose which metrics are available based on inputs to Key Decision Points, then construct the monitoring dashboard that best serves the priorities of each stakeholder.*

Exhibit 7.1. Core Metrics, System *[all model profiles]*

Metric (code)	Primary?	Cadence	Alert threshold, severity	Audience
Scoring availability (SYS-01)	Yes	Per run; daily review	**W** below *[98%]* for 2 days; **C** below *[90%]*	*[Informatics / IT analyst]*
Input record volume (SYS-03)		Daily	**W** outside ±*[30%]* of baseline, 3 days running	
Scored population volume (SYS-06)		Daily	**W** below *[coverage floor]*, 2 days running	

Metric (code)	Primary?	Cadence	Alert threshold, severity	Audience
[Generative] Generation request success (SYS-09)		Hourly	**W** below *[99%]* for a day; **C** below *[95%]* for an hour	
[Generative; or any hosted model per P8] Model version conformance (SYS-10)		Daily	**C** on any request served by an unapproved version	

Exhibit 7.2. Model Performance Metrics *[predictive models, alert-driven]*

Metric (code)	Primary?	Cadence	Alert threshold, severity	Audience
Alert-episode PPV (PERF-02)	Yes	Monthly, trailing 90 days	**W** below *[floor]*, 2 months running	*[Analytics lead, with clinical lead]*
Encounter sensitivity (PERF-03)	Yes	Monthly, trailing 90 days	**W** below *[floor]*, 2 months running	
Discrimination (PERF-05)		Monthly, trailing 90 days	**W** if upper CI bound falls below baseline, 2 months running	
Input feature drift (PERF-10)		Weekly	**W** at PSI > *[0.10]*, **C** at PSI > *[0.25]*, 2 weeks running	

Exhibit 7.3. Model Performance Metrics *[predictive models, not alert-driven]*

Metric (code)	Primary?	Cadence	Alert threshold, severity	Audience
Operating point (PERF-01) *(or Calibration (PERF-06) if there is no threshold)*	Yes	Monthly, trailing 90 days	**W** if sensitivity or PPV below *[floor]*, 2 months running	
Discrimination (PERF-05)		Monthly, trailing 90 days	**W** if upper CI bound falls below baseline, 2 months running	

Metric (code)	Primary?	Cadence	Alert threshold, severity	Audience
Input feature drift (PERF-10)		Weekly	**W** at PSI > *[0.10]*, **C** at PSI > *[0.25]*, 2 weeks running	
Input feature drift (PERF-10)		Weekly	**W** at PSI > *[0.10]*, **C** at PSI > *[0.25]*, 2 weeks running	

Exhibit 7.4. Model Performance Metrics *[generative models]*

Metric (code)	Primary?	Cadence	Alert threshold, severity	Audience
[Human] Human-rated output quality (PERF-18)	Yes	Monthly sample of *[n]*	**W** acceptable share below *[90%]*, 2 cycles running	*[Clinical lead]*
[Human] Clinically significant error rate (PERF-19)	Yes	Same sample	**W** above *[2%]*; **C** on any error that reached patient care	
[Automated] Automated quality score (PERF-20)	Yes	Weekly	**W** pass rate more than *[5]* points below baseline, 2 weeks running	
[Automated] Judge–human agreement (PERF-21)		Quarterly sample of *[n]*, and at every version change	**W** below *[κ 0.6]*; automated score labelled unvalidated until restored	
[If P5 = no routine review or patient-facing] Safety event rate (PERF-23)		Daily screen; weekly rate	**C** on any confirmed event	

Exhibit 7.5. Adoption & Workflow Metrics *[by model type]*

Metric (code)	Model	Primary?	Cadence	Alert threshold, severity	Audience
Alert burden (ADPT-01)	*[Predictive, alert-driven]*	Yes	Daily, trailing 28 days	**W** above *[ceiling per unit-shift]*, 3 days running	*[Clinical operations / nursing leader]*
Engagement (ADPT-05)	*[Predictive, not alert-driven; Generative]*	Yes unless ADPT-08 is selected	Weekly	**I**	

Metric (code)	Model	Primary?	Cadence	Alert threshold, severity	Audience
User-reported issues (ADPT-06)	*[Generative]*		Continuous; reviewed weekly	**W** on any patient-safety-tagged report	
Draft acceptance / edit rate (ADPT-08)	*[Generative, P6 = captured]*	Yes	Weekly, trailing 28 days	**W** major-edit-or-discard share more than *[10]* points above baseline, 3 weeks running	

Exhibit 7.6. Impact & ROI Metrics

Metric (code)	Primary?	Cadence	Alert threshold, severity	Audience
Primary outcome (IMPT-01): *[outcome, definition reference]*	Yes	Monthly, trailing 90 days	**I** (trend; reviewed quarterly)	*[Quality lead / operational sponsor]*

Monitoring Metric Stratification

- **Strata:** *Library defaults (age band, sex, race/ethnicity, preferred language, payer class, care setting or unit type, site), plus or minus per P11.*
- **Stratified metrics:** *Default: the Model Performance and Adoption & Workflow metrics in the core set, plus the primary outcome.*
- **Minimum denominator:** *[20]* outcome events (label-based metrics) or *[n]* units of analysis (other metrics) per subgroup per window; below that, "insufficient data."
- **Small-cell suppression:** cells below *[11]* are suppressed in any report leaving the analytics team.
- **Disparity flag: W** if a subgroup differs from the overall value by more than *[margin]* with non-overlapping CIs, 2 windows running.

Exhibit 7.7. Quality Review Program *[generative models]*

Item	Value
Sampling frame	*Random sample of outputs from the review period, stratified by [unit / output type] so low-volume groups are represented*
Sample size	*[n] outputs per [month]. State the precision this gives, e.g. a 95% CI half-width of about [±6] points on a 90% acceptable share at N = 100*
Reviewers	*[Number] clinicians from [role/specialty]; each output rated by [1] reviewer, with [10%] double-rated to check reviewer agreement*

Item	Value
Rubric	*Dimensions set by the P4 task type, e.g. accuracy against the source, completeness, absence of unsupported content, organization, tone; each rated [1–5], plus an overall acceptable / not acceptable*
Significant-error definitions	*What counts as a clinically significant error for this task (P4), with examples*
Adjudication	*Who confirms a suspected significant error, and within what time*
Escalation	*Any error that may have reached patient care goes to [safety reporting process] within [24 h]*
Reviewer time	*Estimated [n] hours per reviewer per [month]*

08 Reporting & Review

Monitoring metrics can be rolled up into a more detailed evaluation with additional analysis. In production, Vega Health typically provides periodic reviews of monitoring metrics, with metric performance against confidence intervals, a summary of monitoring alerts, and additional insights. In some instances (e.g., a major change to the model or workflow, persistent monitoring alerts or material changes to performance), Vega Health will recommend a deeper model evaluation.

Artifact	Frequency	Contents	Audience	Prepared by	Reviewed by
Monitoring page	Continuous	Four tabs, primary metrics, alerts	Metric owners	Vega Health Platform	Metric owners at their stated cadence
Monthly monitoring summary	Monthly	Status of every core metric; alerts raised and their AI product; System and Adoption trends; *[Generative]* review results for the cycle	Metric owners, Use Case Sponsor	*[Vega Health]*	*[Use Case Sponsor]*
Quarterly performance review	Quarterly	Rollup of all core metrics over the quarter and trailing year: trends with confidence intervals, stratified results and disparity flags, Impact & ROI against the pre-period (P10), incidents and safety events, baseline	Use Case Sponsor; AI governance committee	*[Vega Health, with the customer analytics lead]*	*[AI governance committee]*

Artifact	Frequency	Contents	Audience	Prepared by	Reviewed by
		confirmations, and recommendations (continue, adjust thresholds, investigate, re-evaluate)			
Annual review of Monitoring Plan	Annually	Whether the metrics, thresholds, owners, and review program are still right; amendments	Approving body	*[Vega Health and Use Case Sponsor]*	Approving body

Vega Health may recommend additional an evaluation or analysis to support change management in the following cases:

Plan amendments. Changes to the core set, thresholds, owners, or review program are made as a new plan version and re-approved by the approving body. Threshold tuning within limits stated here *[e.g. ±10%]* may be approved by the Use Case Sponsor and recorded at the next quarterly review.

Model version changes.

[Predictive] A new model version runs alongside the current one on the same live data for *[n weeks]* before cutover and is compared on the core Model Performance metrics (Framework: *Baselines and re-baselining*).

[Generative] The model, prompt, and configuration versions are pinned (Model version conformance (SYS-10)). Before any change, including a provider's foundation-model update, Vega Health re-runs a fixed review set of *[n]* representative inputs on the old and new versions and has clinicians review the results, using the §5.4 rubric. *[Hosted model: the provider's version-retirement notice period and how it is tracked.]* Cutover requires *[Use Case Sponsor]* sign-off.

Re-evaluation triggers. A focused analysis is started if *[any of]*: a critical alert persists beyond its response time; a warning persists for *[3]* review windows; an unresolved disparity flag lasts 2 quarters; a material change in population, workflow, or data sources; *[for generative models]* a confirmed significant error that reached patient care.

Suspension. *Who can suspend the AI product (e.g. Use Case Sponsor, CMIO, Vega Health for a safety or technical cause), on what grounds (e.g. a sustained critical System failure, a patient-safety event attributed to the AI product, an unapproved model version in use), and how users are notified.*

Retirement. Follows the Framework's decommissioning steps: full monitoring through wind-down, shutoff verification, a post-removal Impact window, and a closeout report.

09 Roles & Responsibilities

> *This section outlines the key roles and responsibilities necessary to support the implementation of a Vega Health Monitoring Plan.*

Exhibit 9.1. Key Roles & Responsibilities

Role	Party	Responsibilities
Use Case Sponsor	Customer	Monitoring Plan approval; quarterly review; amendment and suspension decisions
System metric owner	Customer	Reviews System metrics; escalates to vendor support
Model Performance metric owner	Customer	Reviews Model Performance metrics, with the clinical lead
Adoption & Workflow metric owner	Customer	Reviews adoption metrics; leads workflow fixes
Impact & ROI metric owner	Customer	Reviews outcomes at the quarterly review
Vega Health AI Scientist	Vega Health	Monitoring page; monthly summary; quarterly and annual review preparation
Vega Health Engineer	Vega Health	Remediation of Vega Health Platform, pipeline, and model-serving issues

Vega Health's ability to fully implement a Monitoring Plan relies on dependencies, including but not limited to:

- Data feeds required by the core set stay available, with advance notice of interface or mapping changes.
- *[For predictive models]* Outcome data for label-based metrics (P3), at *[latency]*.
- *[For generative models requiring human review]* Reviewer time per §5.4: *[n]* clinician hours per *[month]*.
- *[For generative models]* A channel for users to report problems with outputs (feeds User-reported issues (ADPT-06)).
- *[If Key Decision Point P10 = pre-period]* Historical outcome data for the pre-period.